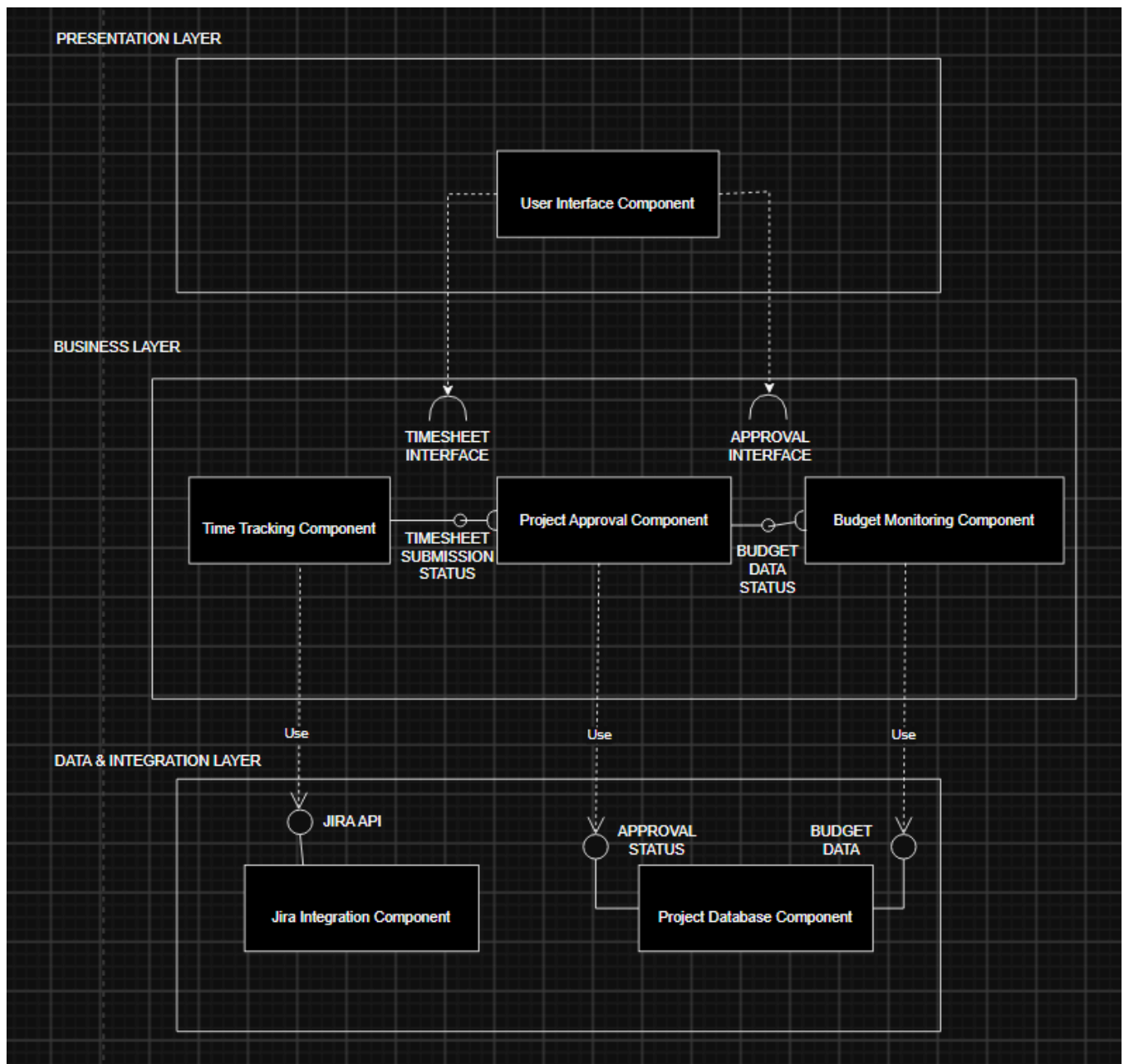


SE BANANA PROBLEM

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DATE : 08/09/2026

1. COMPONENT DIAGRAM



Architecture Selection

Remote Team Time-Tracking & Project Approver

Architecture Selection

For the Remote Team Time-Tracking & Project Approver system, I went with a Layered Architecture.

Reason 1 : Clear Separation of Responsibilities

This approach cleanly splits the system into presentation, business logic, and data/integration layers. That fits the use case well: developers and engineering managers work through the UI, while the actual logic for time tracking, project approvals, and budget oversight lives separately in the business layer, away from what users directly touch.

Reason 2 : Easier Maintenance

Because each core function gets its own dedicated component, the system stays easy to maintain and extend. The Time Tracking Component handles logging developer hours against Jira tickets, the Project Approval Component manages the manager approval workflow, and the Budget Monitoring Component keeps an eye on how fast project budgets are being spent. These map directly to the main functions called out in the problem statement, so each piece can be updated or debugged on its own without disturbing the others.

Security Advantage

By keeping the business logic and data layer separate from the presentation layer, sensitive operations stay isolated from direct user access. The UI never touches project data or approval actions directly. Every request has to go through the proper business component first, which gives a natural checkpoint for enforcing access control.

Performance Benefit

Splitting the system this way also lets frequently used operations, like processing timesheets or tracking budget burn, run independently of everything else. Unrelated functions don't end up competing for the same processing path, which helps keep dashboards responsive even as usage grows.

